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28 June 1988
SUPERSEDING
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18 November 1976

MILITARY SPECIFICATION

SURVEYING SET, ELECTRONIC DISTANCE MEASURING

MICROWAVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, surveying instrument, distance measuring, electronic microwave, herein called SEDME. SEDME units shall be of solid-state construction and are complete with accessories for unsupported operation.

1.2 Part number. The military specification part numbers for the SEDME-MW shall be as follows:

M52156 - W - T

- T: 1, unit supplied with tribrach; 0, unit not supplied with tribrach.
- W: 1, unit supplied with winterization kit; 0, unit not supplied with winterization kit.
- M52156: Document identifier.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

AMSC N/A

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SPECIFICATIONS

FEDERAL

W-C-440	- Clip, Electrical, General Specification for.
QQ-S-781	- Strapping, Steel and Seals.
ZZ-R-765	- Rubber, Silicone: Low- and High Temperature and Tear Resistant.
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-D-950	- Dyeing and Aftertreating Processes for Cotton Cloths.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

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MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-D-3464	- Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-C-4150	- Cases, Transit and Storage, Waterproof and Water-Vaporproof.
MIL-E-4158	- Electronic Equipment Ground; General Requirements for.
MIL-M-10304	- Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-B-11188	- Batteries, Storage: Lead-Acid.
MIL-T-14091	- Tripod, Surveying: Extension-Leg-Type.
MIL-T-14132	- Theodolites, Surveying, Direction: Second Order.
MIL-L-25467	- Lighting, Integral, Aircraft Instrument, General Specification for.
MIL-C-26482	- Connectors, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting.
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible, for Packaging.
MIL-P-55110	- Printed Wiring Boards.
MIL-B-55118	- Batteries, Storage, (Cells), Vented, Nickel-Cadmium.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-681	- Identification Coding and Application of Hookup and Lead Wire.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connnection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 Flair Drive, El Monte, CA 91731.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

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(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The SEDME equipment system offers a distance measuring capability by utilizing two identical units incorporating a microwave source. Distance measurements can be made only by using a pair of identical units in a combination mode. Each of the units of the pair can be operated in either a transmitting or receiving mode of the microwave signal. For a clearer identification of the whole SEDME-LR equipment system as well as the major components of this dual system, special terms will be observed.

SEDME equipment system - is referred to as SEDME and consists of the two complete identical SEDME units in their respective carrying and transport cases, including accessories, tools, and repair kit.

SEDME unit - consists of one half of the pair of units, of the SEDME equipment system, and consists of the following:

numeric control unit (internal or external)
parabolic antenna
horn antenna
electronic package (control unit) containing an X-band solid state microwave source
transport case
psychrometer
altimeter-barometer
meterological correction chart
internal batteries and charger
external nickel cadmium batteries and charger
external power cables (8 meters and 3 meters in length)
spare fuses
tribrach
repair kit
backpack
adaptor (to enable the transmitting instrument to be attached to a tribrach or a tripod via the integral yoke)

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a. SEDME shall measure distances to the required accuracy (see 3.6.3) by transmitting a frequency modulated signal from one SEDME unit, to another SEDME unit and in turn receive back a phase-shift frequency modulated signal. Consequently SEDME shall compute the slope distance, by measuring the phase shift between the two signals, and then display the corrected slope distance in meters or feet after the meteorological data has been entered into the master instrument via a numeric control unit.

b. SEDME shall be capable of measuring in the following configurations:

(1) In a static mode with both instruments stationary and mounted directly on tripods or via tribrachs.

(2) As in (1) but with one or both instruments situated on a mast or tower with its operator.

In all modes the antenna shall be attached directly to the control unit.

The static mode display resolution shall be 1mm (see 3.6.3).

c. The range of the SEDME with the parabolic antenna shall be from 20 meters to 50 kilometers ±5 km, and with the horn antenna from 100 meters to 20 kilometers.

d. The SEDME shall be provided with five microwave channels, selected by the hand held unit, allowing simultaneous operation of two pairs of adjacent SEDMEs.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of SEDMEs from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish SEDMEs conforming to this specification.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more SEDMEs for inspection as specified in 4.4.

3.4 Materials. Materials shall be free from all defects and imperfections that may affect the operation or life of the instrument. Materials used shall be both lightweight and corrosion resistant such as aluminum alloys, magnesium alloys, copper alloys, alloy steel containing a minimum of 11 percent chromium, or molded plastic reinforced with glass fibers.

3.4.1 Material deterioration prevention and control. The SEDME shall be fabricated from compatible materials inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the SEDME may be exposed.

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3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available upon request, to the contracting officer or designated representative.

3.4.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste or reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the SEDME may be newly fabricated from recovered materials to the maximum extent practicable, provided the SEDME produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the SEDME.

3.5 Environmental requirements.

3.5.1 Operational temperature. The SEDME shall operate with no external assistance, without damage from the main and alternate 12-volt power source, with the accuracy specified herein when in thermal equilibrium in air at any temperature from +52 °C (+125 °F) to -18 °C (0 °F). The SEDME shall operate without damage and with the accuracy specified herein from -18 °C (0 °F) to -43 °C (-45 °F) using internal heaters.

3.5.2 Storage temperature. The SEDME shall not be damaged by storage under the following conditions:

Air temperature -	+71 °C (+160 °F)	48 hours
Low temperature -	-57 °C (-70 °F)	24 hours

3.5.3 Solar radiation. The SEDME shall operate, without damage, with the accuracy specified herein when subjected to solar radiation of 1136 watts per square meter (w/m^2) [360 British thermal units per square foot per hour (Btu/sq ft/hr)] for not less than 4 hours in an environmental air temperature of +52 °C (+125 °F), -3 °C (26 °F).

3.5.4 Rain. The SEDME shall not be damaged nor show evidence of water penetration and shall operate within the accuracy specified herein, before and after exposure to windblown [65 Kmph (40 mph) ±8 Kmph (5 mph)] rainfall of 10 cm (4 in.) ±2.5 cm (1 in.) per hour for a period of not less than 2 hours.

3.5.5 Humidity. The SEDME shall not be damaged and shall operate within the specified accuracy when subjected to relative humidities of at least 95 percent at air temperatures varying between +20 °C (+68 °F) ±1 °C (2 °F) and +52 °C (+125 °F) ±1 °C (2 °F).

3.5.6 Sand and dust. When the SEDME transport case is subjected to blowing sand and dust, the SEDME shall not show sand and dust penetration into the electronics package. The accessory case and the transport case shall not show sand or dust penetration.

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3.5.7 Vibration. The SEDME shall not be damaged by vibration nor lose accuracy as specified herein when subjected to the vibration test specified in 4.6.2.12.

3.5.8 Shock. The SEDME shall not be damaged nor lose accuracy as specified herein when subjected to the shock test specified in 4.6.2.13.

3.6 Performance.

3.6.1 Precalibration. Each SEDME shall be calibrated by the contractor. The zero correction value shall be stored in the SEDME microprocessor memory and provision shall be made for automatic change of zero correction when alternative antennas are used.

3.6.2 Compatibility. Each SEDME shall be capable of acting as measurer and responder with any other SEDME built to the same revision of this specification, without any special calibration.

3.6.3 Accuracy and range. When used in a system of two or more instruments, each instrument shall measure distances from 20 meters to 50 kilometers using the parabolic antenna. When using the horn antenna each instrument shall measure distances from 20 meters to 20 kilometers. Each SEDME shall have the following accuracies:

a. In the static mode with the parabolic antenna the probable error for a single determination of distance shall be 10 mm ±3 parts per million of the range except under conditions of high multipath effects. With the horn antenna the probable error for a single determination of distance shall be 30 mm ±3 parts per million of the range except under conditions of high multipath effects. In the dynamic mode using the horn antenna with the instrument moving at relative speeds of up to 30 knots (15 meters per second) the probable error of distance measurement shall be 10 cm.

b. When used in a system of two or more instruments, each instrument shall measure distances from 20 meters to 50 kilometers. Each instrument shall have the following accuracies for the respective range:

20M to 50KM - 1 part in 7,000
5KM to 15KM - 1 part in 25,000
15KM to 50KM - 1 part in 35,000

3.6.4 Warm-up time. The SEDME shall be capable of operation for measurement within the following time limits for the respective temperature ranges:

-18 °C (0 °F) to +52 °C (+125 °F) - No significant warm up time.
-43 °C (-45 °F) to < -18 °C (0 °F) - 10 minutes maximum.

Internal heaters shall be automatically switched on at temperatures below -18 °C (0 °F).

3.6.5 Distance measurement time. After set-up and warm-up of the SEDME, the time required for distance measurement in ambient temperatures from -43 °C (-45 °F) to +52 °C (+125 °F) shall be not more than 20 seconds.

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3.6.6 Automatic tuning. The SEDME shall tune in automatically to either of two microwave channels selected by a control unit, within 100 milliseconds after both instruments, of a SEDME pair, are switched on. Switching of the carrier, modulation frequencies, phase shift measurement and total corrected or uncorrected distance measurement (see 3.1a) shall be done automatically under microprocessor control. The microprocessor shall be used additionally for speed memory and smoothing of range data in the dynamic mode. It shall also be used for self testing purposes.

3.6.7 Pattern determination. A modulation frequency of 150 MHz shall be used for the "finest" pattern (*); to enable single shot measurement on one carrier frequency to be achieved.

3.7 Physical characteristics.

3.7.1 Portability. The SEDME equipment shall be furnished in two soldier-portable packages each of which contains a SEDME unit.

3.7.2 Weights. The SEDME equipment packages shall not weigh more than the following:

Item	Maximum Weight Kilograms	(Pounds)
Control unit incorporating microwave source and integral yoke	4.5	(10.0)
Control unit	0.8	(1.75)
Altimeter-Barometer	0.9	(2.0)
Backpacks	3.0	(6.5)
Cable for alternate power source	2.7	(6.0)
Cable for main power source	0.4	(1.0)
Main power/internal source battery pack	1.4	(3.0)
Psychrometer (sling type)	0.91	(2.0)
Parabolic antenna	1.5	(3.3)
Horn antenna	0.4	(0.88)

3.7.3 Volume and dimensions.

a. The volume of the control unit, excluding yoke, shall not be more than 0.007 cubic meters (0.25 cubic feet). The dimensions shall be approximately 260 mm (10.2 inches) deep x 200 mm (7.9 inches) wide x 135 mm (5.3 inches) high.

The overall dimensions of the control unit including yoke shall not be more than 260 mm (10.2 inches) deep x 200 mm (7.9 inches) wide x 135 mm (5.3 inches) high.

* Ability to interpret the wave form with a high-degree of intelligence.

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b. The volume of the parabolic antenna shall not be more than 0.005 cubic meters (0.18 cubic feet). The diameter shall not be more than 340 mm (13.4 inches) and the height shall not be more than 170 mm (6.7 inches). The volume of the horn antenna shall not be more than 0.0006 cubic meters (0.022 cubic feet) and the dimensions shall not be more than 110 mm (4.3 inches) x 110 mm (4.3 inches) aperture x 140 mm (5.5 inches) long.

3.7.4 Controls. The control unit and SEDME dials, knobs, and switches shall be grouped functionally and in accordance with the human factors engineering requirements of 3.21 and MIL-E-4158. Dials, knobs, and switches shall be protected and shall be compatible for use by personnel wearing arctic mittens. The layout shall be submitted to the contracting officer for approval of control location and operation in regard to human factors engineering.

3.7.5 Control unit. The control unit shall provide a digital readout of the corrected or uncorrected slope distance in meters with a resolution of 1 centimeter. The numerical characters shall not be less than 6.35 mm (.25 inch) high.

3.7.6 Receptacles. SEDME receptacles shall conform to MIL-C-26482. One receptacle shall be for the external power supplies (main power supply, if applicable and the alternate power supplies). A separate receptacle shall be for the headset microphone. The receptacles shall be labeled POWER SUPPLY and HEADSET, respectively.

3.7.7 Radio location band. The SEDME microwave transmission shall be between 15.7 Gigahertz and 17.7 Gigahertz.

3.8 Electronics package.

3.8.1 Modules. The electronics package shall be designed and arranged in modular units and shall have a positive guide arrangement to facilitate module removal and replacement.

3.8.2 Captive hardware. Captive hardware shall be used on all screws that can be loosened from the exterior of the electronics package.

3.8.3 Component boards. The components shall be mounted on printed wiring boards in accordance with MIL-P-55110. The plug-in component boards shall be equipped with a coded keying arrangement.

3.8.4 Panel marking. Markings on the control panel shall identify each control, meter, and connector by its functional name, symbol, or operating instruction reference.

3.8.5 Panel illumination. All operating indicators and dial illumination shall be adjustable and controlled by a front panel knob. The color shall be as specified in MIL-L-25467. All lamps except for meters shall be replaceable from the outside of the electronics package.

3.8.6 Electronic components and parts. Electronic components and parts shall conform to MIL-E-4158 or alternatives which are acceptable within the total environmental performance of the instrument. No hand-selected components shall be used.

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3.8.7 Color coding. Hookup and chassis wiring shall be color coded in accordance with MIL-STD-681.

3.8.8 Test meter. A built-in test meter in accordance with MIL-M-10304 shall be provided for monitoring electrical circuits as necessary to ensure the SEDME equipment is operating normally. When used in conjunction with the maintenance manual, the test meter shall give an indication as to which module or subassembly is not functioning properly insofar as this is practical. The test meter may have internal illumination in accordance with MIL-L-25467.

3.8.9 Signal strength meter: This requirement is covered by the functions of the test meter as stated in 3.8.8.

3.8.10 Secure lighting. Ninety-five percent of all external lighting energy, emitted between 300 and 1100 nanometers, shall be a blue-green color between 400-700 nanometer wave lengths. No light shall be visible beyond a 50 meter circle with the instrument as central point under moonless and starless night conditions. All external visible light produced shall be intensity adjustable and shall not exceed current level footlambert intensity.

3.9 Power. The SEDME shall operate on not more than 24 watts with the power sources listed in 3.9.1 and it shall operate in the voltage range of 11.0 to 34.0 volts D.C. SEDME shall contain built-in protection against reverse polarity from the power source. At temperatures below -18 °C (0 °F) the additional power required for internal heating shall not exceed 10 watts.

3.9.1 Power sources.

3.9.1.1 Main power sources. The battery shall be a nickel-cadmium type conforming to MIL-B-55118 with a combination of cells to give 12-volt operation. The battery shall be rechargeable via a 110/220-volt, 60 Hz charger supplied with the SEDME and shall provide power sufficient for 2 hours operation. The battery(s) shall be spillproof and shall have no detectable liquid or gas regardless of operation or storage orientation and shall not be damaged when subjected to shock testing as specified in 3.5.8. The battery shall not be damaged nor lose capacity when charged continuously from the charger supplied with the system.

3.9.1.1.1 External power source. When the main power source is an external battery, it shall be attached to the SEDME electronics package via a power cable (see 3.10.1) with quick-disconnect receptacles.

3.9.1.2 Alternate power source. The SEDME shall operate from batteries conforming to MIL-B-11188, 12-volt D.C., class FE for 4 hours and also from vehicle batteries mounted in a military truck.

3.9.1.3 Terminal covers. Power sources shall be supplied with terminal covers.

3.10 Power cables. One end of each cable shall terminate in a plug conforming to MIL-C-26482, quick-disconnect, straight or angle, size and arrangements as required, which shall fit the equipment receptacle.

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3.10.1 Main power cable. If the main power source is from an external battery, a cable at least 3 meters (10 feet) long shall be supplied. Each end of the cable shall terminate in a plug conforming to MIL-C-26482, quick-disconnect, straight, or angle, size and arrangements as required, one of which will fit the battery output receptacle, the other to the SEDME equipment.

3.10.2 Alternate power cable. The alternate power cable shall be at least 8.0 meters (26.2 feet) long and shall terminate in battery clips conforming to W-C-440, type PC, with insulators.

3.10.3 Wire and cable marking. Mechanically separable connections shall be marked indelibly with characters of a size and color to ensure legibility. Cables shall be identified at the break-out of each branch and at each termination. Wire within cables shall be marked individually at each termination.

3.11 Headset-microphone. A volume/loudness adjustable headset-microphone shall be furnished with each SEDME equipment.

3.12 Cases.

3.12.1 Control unit case. The construction of the control unit case shall be in accordance with MIL-C-4150 except for the drop and immersion provisions. The control unit case shall serve as the field carrying and storage case and shall protect the electronics package against shock, vibration, and environmental conditions as specified herein. The entire case shall be dust and moisture tight. All gaskets shall be in accordance with ZZ-R-765, class 1a, 50 durometer hardness. A carrying handle shall be fitted which is compatible for use with arctic mittens. Provision shall be made for mounting the control unit directly and indirectly via tribrach on a tripod as specified in 3.14 using a 5/8-11 UNC thread. Threads shall be in accordance with FED-STD-H28.

3.12.2 Transport case. Each of the two SEDME units shall be provided with a transport case conforming to MIL-C-4150. Provision shall be made for storage of the carrying packs (see 3.15) in the transport case. Cushioning material in accordance with MIL-P-26514 shall be provided to protect the contents against shock, vibration, and environmental conditions as specified herein during shipment and storage. Cut-outs or special space shall be provided to accommodate any anticipated accessories, tools, and repair kit.

3.13 Winterization kit. When specified (see 6.2), a winterization kit shall be furnished for use with SEDME at temperatures below -18 °C (0 °F) (see 3.6.4). The winterization kit shall include a cover or covers for all controls, switches, dials, and meters and a canvas or cotton duck bag for carrying and storing the winterization kit. The winterization kit shall also operate from the alternate power sources as specified in 3.9.1.2.

3.14 Tripod. Tripods conforming to MIL-T-14091 shall be furnished with wrench, square-point screwdriver, and plumb bob (which shall weigh at least 0.34 kg (12 oz)) (see 6.2).

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3.15 Backpacks. Means for transporting the SEDME by backpacking shall be provided. The backpacking shall be human engineered for use by 5th percentile female through 95th percentile male soldiers and shall comply with requirements of 3.21 for portability and load carrying. Canvas, if used, shall conform to CCC-C-419, type and weight to suit the purpose, and water-repellent in accordance with CCC-D-950, type I, class B, inhibitor f, and the color shall be forest green as specified in MIL-T-704. There shall be two backpacks, one for the equipment and one for the accessory case.

3.16 Adapter unit. The adaptor unit shall allow the instrument to be mounted on a tribrach via the instrument yoke which allows the instrument to be adjusted for maximum return signal in the azimuthal and vertical planes.

3.17 Tribrach. When required, tribrachs conforming to MIL-T-14132 with an optical plummet shall be furnished as specified in 6.2.

3.18 Psychrometer. The psychrometer shall be a sling type for measuring wet and dry bulb temperatures. The instrument shall be furnished with a thermometer readable in both ± degrees Celsius and ± degrees Fahrenheit.

3.19 Altimeter-barometer. The altimeter-barometer shall be an aneroid type that shall measure the barometric pressure in millibars of mercury, be temperature compensated and be able to convert the barometric pressure to altitude in meters. It shall have a resolution of at least 3 millibars for the barometric pressure and at least 30 meters for the altitude reading. It shall not be more than 8 centimeters (3.2 inches) in diameter and 5 centimeters (2.0 inches) deep. It shall not weigh more than 0.4 kilogram (1 pound).

3.20 Accessories and tools. The following shall be the minimum accessories and tools furnished for each individual SEDME unit:

Item	Quantity
Adapter unit	1
Altitude-barometer, millibars of mercury (Hg)	1
Backpack	1
External power cable	1
Instruction manual(s)	1 each
Meteorological correction chart (metric)	1
Psychrometer, sling and thermometer(s)	1 each
Repair kit	1
Screw driver (square point)	1
Spare fuses	5
Tribrach	1
Tripod	1
Wrench	1

3.21 Human factors engineering. The SEDME shall conform to the human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, General Requirements (4), Control/Display Integration (5.1), Visual Displays (5.2), Audio Displays (5.3), Controls (5.4), Labeling (5.5), Anthropometry (5.6), Design for Maintainability (5.9), Small Systems and Equipment (5.11), and Hazards and Safety (5.13) of MIL-STD-1472, as applicable.

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3.22 Reliability. The specified Mean-Time-Between Failure (MTBF) of the SEDME shall be at least 500 hours as tested in 4.6.2.17.

3.23 Maintainability.

3.23.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, and test equipment, as specified in 6.6 and 6.7, shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in section 4.

3.23.2 Maintenance ratio. The SEDME shall have a maintenance ratio of not more than 0.05 when tested as specified in 4.6.2.18.2. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.23.3 Ease of maintenance. The SEDME construction shall provide ease of access to batteries and fuses (or reset buttons) such that tools are not required to connect a battery or external power nor to change a fuse. The equipment construction shall also provide for a self-test display verifing satisfactory operating capability prior to an initial distance measurement. The SEDME shall comply with the Design for Maintainability requirements of 3.21.

3.24 Electromagnetic interference. The SEDME shall be tested in accordance with MIL-STD-462, methods CE0 3 for conducted emissions and RE0 2 for radiated emissions and to the requirements of MIL-STD-461, figures 14 and 22.

3.25 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II as applicable (see 3.25.1), with 1 percent copper 8-quinolinolate (by weight) based on the non-volatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.26 Identification marking. The SEDME shall be identified in accordance with MIL-STD-130. The marking shall be applied to the applicable parts of the SEDME on labels conforming to MIL-P-514, type I, style 2 composition A, class 2 material. Plates shall be attached by adhesive binder in a location where the plate will be both visible and legible.

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3.27 Treatment and painting. The portions of the SEDME normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G. Unless otherwise specified (see 6.2), the color shall be green 383, color number 34094. Nonpermissible galvanic couples, as defined in MIL-T-704, type A, shall be insulated as defined in MIL-T-704, type A.

3.28 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance. The interchangeability of printed circuit boards shall be such that there is no detriment to other circuitry and no realignment is required to maintain the specified functions of the instrument.

3.29 Workmanship. All parts, components, and assemblies of the SEDME equipment tripod, and cases including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All parts shall be finished. There shall be no cracks, sharp bends, dents, or chips in finished parts. There shall be no loose solder or insecure electrical connections. The fit of assemblies and sub-assemblies shall be made without imposing strain or forcing parts. All threads requiring movement and all sliding and rotatable parts shall move without binding, backlash, or endplay. Clamping actions, when required, shall be positive and, when clamped, shall not cause drag. Bearing surfaces shall show no scratches or any irregular surface blemishes. External surfaces shall be free from burrs and sharp edges, and corners shall be chamfered and rounded except when sharp edges and corners are required.

3.30 Safety. The SEDME shall present no uncontrolled hazards to operators, maintenance personnel or bystanders and shall conform to all applicable requirements of Requirement 1 of MIL-STD-454. A dummy load shall be provided so that all microwave energy will be absorbed when maintenance of the unit requires the unit to be operated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all

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products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified here are classified as follows:

a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 Examination. The first article SEDME shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article SEDME shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production SEDME equipment will be selected at random by the Government from the equipment being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in column 2 of table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production SEDME shall not exclude the remaining instruments from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production SEDME to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production SEDME and shall be cause for refusal by the Government to continue acceptance of production instruments until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on any SEDME previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all SEDME equipment

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accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Samples. Samples selected in accordance with 4.5.1 shall be examined for the major and minor defects specified in 4.6.1. The acceptable quality level (AQL) shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in column 3, table I. AQL shall be 1.0 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The SEDME shall be examined as specified herein for the following defects:

Major

101. Operating frequency band not as specified.
102. Material not as specified.
103. Material deterioration prevention and control not as specified.
104. Dissimilar metal not as specified.
105. Identification of materials and finish not as specified.
106. Recovered material not as specified.
107. Parts, components, or subassemblies not as specified.
108. Controls not as specified.
109. Illumination not as specified.
110. Read-out not as specified.
111. Batteries not as specified.
112. Accessibility of parts and components for service and maintenance not as specified.
113. Dimensions and weights not as specified.
114. Test points not provided for electrical circuitry.
115. Winterization kit not as specified.
116. Power source not as specified.
117. Backpack portability not as specified.
118. Cases not as specified.
119. Instrument modules not as specified.
120. Safety not as specified.

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Minor

201. Treatment, finishing, or painting not as specified.
202. Workmanship not as specified.
203. Identification or special marking missing or illegible.
204. Technical publications missing (when required).
205. Fuses not as specified.
206. Instrument seals against moisture not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Prior to beginning the tests, the contractor shall prepare the SEDME to operate as specified herein. The test chamber used for the environmental tests shall conform to MIL-STD-810. All tests shall be conducted in sequence indicated in table I.

4.6.2.2 Test schedule.

TABLE I. Test schedule.

First article	Initial production	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	X	X	Examination	4.6.1	3.1
X	X	X	Accuracy, range, time	4.6.2.3	3.6.3
X	X	X	Compatibility	4.6.2.4	3.6.2
X	X	-	High temperature storage	4.6.2.6.1	3.5.2
X	X	-	Performance	4.6.2.5	3.6
X	X	-	High temperature operation	4.6.2.6.2	3.5.1
X	X	-	Performance	4.6.2.5	3.6
X	X	X	Low temperature	4.6.2.7	3.5.2
X	X	X	Performance	4.6.2.5	3.6
X	X	-	Solar radiation	4.6.2.8	3.5.3
X	X	-	Performance	4.6.2.5	3.6
X	X	-	Rain	4.6.2.9	3.5.4
X	X	-	Performance	4.6.2.5	3.6
X	X	-	Humidity	4.6.2.10	3.5.5
X	X	-	Performance	4.6.2.5	3.6
X	X	-	Sand and dust	4.6.2.11	3.5.6
X	X	-	Performance	4.6.2.5	3.6
X	X	X	Vibration SEDME in transport case	4.6.2.12.1	3.5.7
X	X	X	Performance	4.6.2.5	3.6

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TABLE I. Test schedule. (Cont'd)

First article	Initial production	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	X	-	Vibration SEDME out of transport case	4.6.2.12.2	3.5.7
X	X	-	Performance	4.6.2.5	3.6
X	X	-	Shock SEDME in transport case	4.6.2.13.1	3.5.8
X	X	-	Performance	4.6.2.5	3.6
X	X	-	Shock SEDME out of transport case	4.6.2.13.2	3.5.8
X	X	-	Performance	4.6.2.5	3.6
X	X	-	Voltage and power consumption	4.6.2.14	3.9
X	X	-	Electromagnetic interference	4.6.2.15	3.24
X	X	-	Human factors engineering	4.6.2.16	3.21
X	X	-	Reliability	4.6.2.17	3.22
X	X	-	Maintainability	4.6.2.18	3.23
X	X	-	Accuracy, range, time	4.6.2.3	3.6.3, 3.6.4
X	X	-	Safety	4.6.2.19	3.30

4.6.2.3 Accuracy, range, and time. The SEDME shall be tested for accuracy, range, and time, as pairs, in the following setups:

a. A baseline consisting of two monuments which are at the same altitude (±0.5 meter) and 210 (+10, -0) meters apart; a 10-meter beam graduated in meters with one end of the beam aligned vertically over one monument, the other end of the beam aligned between both monuments and the slope of the beam not greater than 0.05 shall be used. Each of the two SEDME equipments shall be positioned vertically over their respective monument or meter graduation, within 0.5 meter of each other. Measurement shall be made at each meter graduation on the beam. An error between the difference of the consecutive measurements and 1 meter greater than 0.03 meter or the time to complete the measurements not conforming to 3.6.4 shall constitute failure.

b. A baseline consisting of a master monument and four station monuments in line; at the same altitude (±0.5 meter); with distances ranging from 200 to 900 meters (+10, -0 meters) between the master and station monuments; with the station monuments 150 to 300 meters apart and the horizontal distances having an accuracy of 1 part in 35,000, shall be used. A measurement shall be made between the

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master and each station monument with the corresponding time noted. Accuracy and corresponding time not conforming to either 3.6.3 or 3.6.4, respectively, shall constitute a failure.

c. A baseline consisting of a master monument and remote monument located 50 kilometers (+5 kilometers apart) with a horizontal accuracy of 1 part in 35,000 shall be used. Each SEDME shall be positioned over its respective monuments, a series of three distance measurements taken, and corresponding times noted. Accuracy and corresponding time measurements not conforming to 3.6.3 and 3.6.4, respectively, shall constitute failure of the test.

4.6.2.4 Compatibility. The SEDME shall be tested against one another in pairs as specified in 4.6.2.3.

4.6.2.5 Performance. The SEDME shall be inspected for accuracy maintainability prior to and after each mechanical and environmental test specified in table I. The baseline used for this inspection may be any of the baselines mentioned herein or may be a reference baseline established at the test site. The test shall consist of aligning pairs of SEDME over the monuments, taking a distance measurement, and noting the time. Accuracy and corresponding time measurements not conforming to 3.6.3, 3.6.4, and 3.6.5 shall constitute failure of the test.

4.6.2.6 High temperature.

4.6.2.6.1 Storage. Procedure I of method 501.1 in MIL-STD-810 shall be followed. Nonconformance to the requirements of 3.6.3 through 3.6.5 shall constitute a failure.

4.6.2.6.2 Operation. Procedure II of method 501.1 in MIL-STD-810 shall be followed except in step 3 a measurement shall be made. Nonconformance to the requirements of 3.6.3, 3.6.4, and 3.6.5 shall constitute a failure.

4.6.2.7 Low temperature. Method 502.1 in MIL-STD-810 shall be followed. Nonconformance to the requirements of 3.6.3 through 3.6.5 shall constitute a failure.

4.6.2.8 Solar radiation. Procedure I of method 505.1 in MIL-STD-810 shall be followed except that the duration of the test shall be 4 hours and interim measurements shall be made every half-hour during the test for a total of eight measurements. Nonconformance to the requirements of 3.6.3 through 3.6.5 shall constitute a failure.

4.6.2.9 Rain. With the SEDME in an operational configuration it shall be subjected to a simulated rain at the rate of 10 +2 cm per hour for 2 hours consisting of four 30-minute periods. During a 30-minute period, a rainfall of 10 +2 cm per hour shall be simulated; a wind source producing a horizontal wind velocity of 65 km per hour shall be turned on for the first 15 minutes after which it shall be turned off; a protective cover may be employed to protect the SEDME from the rainfall after the first 5 minutes, for 25 minutes out of each 30-minute period. Each face of the SEDME, the accessory case, and the transport case shall be pointed into the wind source during a 30 minute period. After the 2-hour test period is completed, the SEDME and the cases

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shall be checked for evidence of damage to operating parts and penetration of moisture into the electronics package, antenna, and cases, which shall constitute a failure of the test.

4.6.2.10 Humidity. Procedure II of method 507.1 in MIL-STD-810 shall be followed except that only one 48-hour cycle of figure 507.1-2 shall be used. Nonconformance to the requirements of 3.6.3 through 3.6.5 shall constitute a failure.

4.6.2.11 Sand and dust. Method 510.1 in MIL-STD-810 shall be followed. Nonconformance to the requirements of 3.5.6 shall constitute a failure.

4.6.2.12 Vibration.

4.6.2.12.1 SEDME in transport case. The SEDME with all accessories shall be packed in the transport case and subjected to procedure X of method 514.2 in MIL-STD-810 using test levels of curve "AX" in figure 514.2-VIII. The SEDME and accessories shall be checked for physical damage. Nonconformance to the requirements of 3.5.7 shall constitute a failure of this test.

4.6.2.12.2 SEDME out of transportation case. The SEDME shall be mounted to the vibration table. They shall be vibrated along each of three orthogonal axes for a period of 1 hour, through a frequency range of 10 to 55 Hz, with a 0.25 cm double amplitude or maximum acceleration of 1.0g with a sweep time of 15 minutes logarithmically. The sweeptime includes an ascending plus a descending sweep of the frequency range. The SEDME equipment and accessories shall be checked for any physical damage, which, if it does exist, shall constitute a failure of this test.

4.6.2.13 Shock. Both the transport case and the SEDME shall be fully loaded when tested during the shock test in 4.6.2.13.1.

4.6.2.13.1 SEDME equipment system in transport case. The transport case shall be subjected to procedure II of method 516.2 in MIL-STD-810. At the conclusion of this shock test, SEDME equipment system shall be inspected for damage. Evidence of damage (see 6.4), to SEDME equipment componentry, excluding cases, shall constitute failure of the test. Damage to cases shall be allowed provided that it does not prohibit the case from continuing its mission. Examples are as follows:

a. Superficial hairline cracks;
b. Deformation of sealing edges less than 0.32 cm end-to-end of case which does not inhibit action, leveling action, or sealing;
c. Wrinkling of corners which will rebound;
d. Locking latches distorted but not loose at mountings or allowing case seal to deteriorate and still able to function;
e. Handles abraded but not deformed; case abrasion but no through-the-wall rupture.

4.6.2.13.2 SEDME equipment system out of transport case. The SEDME equipment out of the transport case shall be tested according to procedure V of method 516.2 in MIL-STD-810. After the test, the SEDME shall be checked for physical damage as described in 6.4. If any damage exists, a test failure has occurred.

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4.6.2.14. Voltage and power consumption.

4.6.2.14.1 Voltage range. The SEDME shall be set up on any of the baselines used for the accuracy test, with an adjustable voltage having an output voltmeter as the power source. SEDME shall then be operated with the output voltage set at the following levels: 11, 14, 22, 25, 28, and 34. The accuracy of the measurements at each of the voltages previously defined shall be checked for conformance to 3.6.3. Nonconformance to 3.6.3 shall constitute a failure.

4.6.2.14.2 Power consumption. SEDME shall be set up on a baseline with a voltmeter and ammeter or power meter to monitor the instrument's power consumption during a measurement and warm-up. Nonconformance to the requirements of 3.9 shall constitute a failure.

4.6.2.14.3 Battery life.

4.6.2.14.3.1 Main power source. The SEDME shall be set up on a baseline used in 4.6.2.3 and measure the distance using the main power source (NiCad battery), until the power is insufficient for operation of the SEDME equipment or the accuracy of the measurements are not in compliance with 3.6.3. The total operating time shall be determined. Nonconformance to 3.9.1.1 shall constitute a failure.

4.6.2.14.3.2 Alternate power source. The SEDME shall be set up on a baseline used in 4.6.2.3 and the distance measured using the alternate power source (lead acid battery), until the power is sufficient for operation of the SEDME equipment or the accuracy of the measurements are not in compliance with 3.6.3. The total operating time shall be determined. Nonconformance to 3.9.1.2 shall constitute a failure.

4.6.2.15 Electromagnetic interference. Methods CE 03 for conducted emissions and RE02 for radiated emissions in MIL-STD-462 and MIL-STD-461, figures 14 and 22 shall be used. Nonconformance to MIL-STD-462 and MIL-STD-461 shall constitute a failure.

4.6.2.16 Human factors engineering. The SEDME shall be tested for human factors engineering as required in 3.21. During the course of all testing herein, test directors shall scrutinize the SEDME equipment to determine hardware compatibility to required operator functions. The SEDME shall also be evaluated by a government Human Factors Engineer for conformance to 3.21. The following functions shall be noted specifically:

a. Suitability for backpack.
b. Operational use with artic mittens.
c. Adaptability to night operation.
d. Location and configuration of all controls.
e. Compatibility of operation with other surveying hardware.
f. Adaptability to battery charging.
g. Interchangeability of SEDME componentry with other hardware during operation.
h. Adaptability to easy "setup" in field operations.

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i. Hardware presents no safety hazard from electronics, or mechanical hazard to operator, maintenance personnel, and others.

Nonconformance with the requirements of 3.21 shall constitute failure of this test.

4.6.2.17 Reliability. Using an MTBF of 500 hours, SEDME shall be tested in accordance with MIL-STD-781, test plan VI. The SEDME shall be placed in an environmental chamber in the operational mode with a chamber temperature of 55 °F for the duration of this test. Power to the instruments shall be from an external source. Measurements shall be initiated and continued for the duration of this test with the following exceptions:

(1) The instruments shall be cycled through "self-check" twice each minute.

(2) The instruments shall be turned off (using the instrument switch) for a period of 30 ±5 seconds once in each ten minute period. This procedure will continue until a decision can be made from test plan VI. Inability of an instrument to perform any of the operational functions shall be considered a failure.

If an "accept" decision is made the instruments shall be removed from the chamber and tested in accordance with 4.6.2.5. Reaching a "reject" decision or nonconformance with 3.6.3, 3.6.4, or 3.6.5 shall constitute failure of the reliability test.

4.6.2.18 Maintainability.

4.6.2.18.1 Maintenance support evaluation. Maintenance operation required during initial production testing shall be accomplished to determine conformance to 3.23. Failure of the support elements to maintain the SEDME equipment as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio.

4.6.2.18.2 Maintenance ratio. The maintenance ratio shall be computed during testing specified in 4.6.2.17. Operating time to perform each mission and the active maintenance man-hours required for both scheduled and unscheduled maintenance shall be recorded. Nonconformance to 3.23.2 shall constitute failure of this test.

4.6.2.19 Safety. Microwave radiation levels emanating from the equipment shall be verified and documented. 360° measurements shall be made around the equipment to determine if there are any areas where unintended radiation exists. Leakage test shall be performed to determine if the dummy load as specified in 3.30 performs its intended function. Unintended radiation levels above 5 MW/CM^2 at 5 CM shall constitute failure of the test.

4.7 Inspection comparison. The Government's contracting officer may select any SEDME equipment at any time during the contract production period and subject these SEDMEs to the examination specified in 4.6.1 and the tests marked "X" in column 3 of table II in order to determine conformance to the requirements of this specification. The inspection will be performed by the

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Government's representatives, at a site selected by the Government, on SEDMEs selected at random from those which have been accepted previously by the Government but will not include the previously inspected first article and SEDME initial production. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any or all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison SEDME equipment to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison SEDME equipment and shall be cause for refusal by the Government to continue acceptance of the production SEDME equipment until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on SEDME equipment previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all SEDME equipment accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of a pack.

4.8.1.1 Unit of product. For inspection purposes a completed pack prepared for shipment shall be considered a unit of product consisting of the whole SEDME equipment system.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Pack samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

Major defects

121. Desiccant not placed in the case as specified for level A or C.
122. Quantity of desiccant not in accordance with the referenced formula as specified for level A or C.
123. Wood block not secured in end of the box containing the tripod as specified for level A or C.
124. Fiberboard boxes not closed and secured in accordance with the referenced method as specified for level A or C.
125. Strapping not as specified for level A, B or C.
126. Marking missing, illegible, incorrect, or incomplete for level A, B, or C.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. All components for which cases are provided shall be placed in their respective cases and the case lids closed and secured. In addition, the case containing the SEDME equipment system shall have desiccant placed within the case. The desiccant shall conform to MIL-D-3464 and shall be in the quantity required in accordance with the formula specified in MIL-P-116. The components shall be placed in the transport case as provided for and the case lid closed and secured. Components such as the tripod, for which cases are not provided, shall be placed in close-fitting boxes conforming to PPP-B-636, W5c, style optional. The box containing the tripod shall have a fitted block of wood secured in the end of the box to prevent the tripod legs from puncturing the box. Each box shall be closed and secured in accordance with method V of the appendix to the box specificiation.

5.1.2 Level C. Preservation shall be as specified in 5.1.1 for level A except the boxes for the components without cases shall be domestic fiberboard.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. The components comprising a complete SEDME equipment system, including those in carrying cases, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional. The contents shall be blocked or braced within the box in a manner to prevent movement or damage. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A except that the box shall be domestic type and strapping shall be finish A.

5.2.3 Level C. The components comprising a complete SEDME equipment system, cased and uncased, preserved as specified in 5.1, shall be packed in domestic fiberboard shipping containers of a type appropriate for the size and weight of the contents. The boxes shall be strapped with finish A strapping to assure carrier acceptance and safe delivery to the contracting officer's specified destination at the lowest ratings in compliance with the Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. A minimum of two SEDME units one operating in the measuring mode and the other operating in the answering mode, are intended for the precise measurement of survey lines.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of first article, and number of SEDMEs required (see 3.2).
c. When the Government will conduct any or all of the production model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the contractor (see 3.2).
d. When the initial production inspection is required and number of SEDME equipment to be furnished (see 3.2).
e. Winterization kit is required (see 3.13).
f. Level of preservation and packing required (see 5.1 and 5.2).
g. Quantity of tripods as required. See 3.14.
h. Quantity of tribrachs as required. See 3.17.
i. Quantity of thermometers per MIL-T-12625 to be determined at contract time.

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of 2 or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Definition. The following definition shall apply throughout this specification.

6.4.1 Damage. Damage is deformation, corrosion, loosening of parts, breakage, or change of fit of any component or part, or any physical condition resulting in noncompliance of the SEDME equipment system with provisions of this specification.

6.5 Safety. The contracting officer should arrange to furnish to the contractor a copy of TB MED 523 and require the contractor to furnish certified electromagnetic radiation data of the microwave output, of a representative surveying instrument, of his own manufacture that complies with this specification to the US Army Environmental Hygiene Agency in accordance with AR 40-5, for the purpose of determining that all applicable safety requirements specified in 3.21 have been complied with.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each SEDME. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management Control District.

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6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the SEDME.

6.8 Subject term (key word) listing.

Microwave source

Parabolic antenna

Psychrometer

SEDME

Slope Distance

Tribrach

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 99

Review activities:
Army - ER, CE
DSA - GS
Air Force - 82

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 6675-0833

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-52156D	Surveying Set, Electronic Distance Measuring Microwave

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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